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UNIVERSITY OF NAIROBI

BSC. Computer Science Year 1. SMA3211 Linear Algebra 2025/2026

Continuous Assessment Test (C.A.T)

Date: April 24, 2026. Time: 12:00 PM - 13:00 PM

Answer all the questions in the spaces provided.

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1. The system of equations

$$x + y + z = 6$$
$$x + 4y + 6z = 20$$
$$x + 4y + \lambda z = \mu$$

has NO solution for values of λ and μ given by;

(a) $\lambda = 6, \mu = 20$ (b) $\lambda \neq 6, \mu = 20$ (c) $\lambda = 6, \mu \neq 20$ (d) $\lambda \neq 6, \mu \neq 20$

Justify! (5 MARKS)

0

~~Augmented matrix~~

~~det(Coefficient matrix)~~

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 4 & 6 \\ 1 & 4 & \lambda \end{pmatrix} \begin{matrix} R_2 - R_1 = R_2 \\ R_3 - R_1 = R_3 \end{matrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 3 & 5 \\ 0 & 3 & \lambda - 1 \end{pmatrix} R_3 - R_2$$

~~det(Coefficient matrix) = 0~~

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & \lambda - 6 \end{pmatrix}$$

~~det(Coefficient matrix)~~

$\det(A) = 0$ since the system has NO solution

$\therefore 1 \times 3(\lambda - 6) = 0$

$3\lambda - 18 = 0$

$\frac{3\lambda}{3} = \frac{18}{3}$

$\lambda = 6$.

$x + 4y + 6z = 20$

$x + 4y + \lambda z = N$ -

$(6 - \lambda)z = 20 - N$

but $\lambda = 6$

$\therefore (6 - 6)z = 20 - N$

$0z = 20 - N$

$0 = 20 - N$

$N = 20$

$\therefore \lambda = 6$ and $N = 20$.

2 The system of linear equations

$$\begin{pmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 14 \end{pmatrix}$$

has:

(a) A unique solution (b) Infinitely many solution (c) No solution (d) Exactly two solutions.

And state whether the system is consistent or inconsistent. Inconsistent.

Reason: (5 MARKS)

$A = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{pmatrix}$

using the second row,

$\det(A) = -3\{(5\times1) - (2\times3)\} - 1\{(2\times2) - (1\times1)\}$

$\det(A) \neq 0$

$= -3(5-6) - 1(4-1)$

$= -3(-1) - 1(3)$

$= 3 - 3$

$= 0$

∴ The system of linear equations has no solution.

∴ The system of linear equations is inconsistent.

0

3 For matrices of the same dimension M, N and scalars c, which of these properties DOES NOT ALWAYS hold!

(a) $(M^T)^T = M$ (b) $(cM)^T = c(M)^T$ (c) $(M+N)^T = M^T + N^T$ (d) $MN = NM$

Clearly Show!(5 MARKS):

Let $N = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$

$MN = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$

$NM = \begin{pmatrix} ae+cf & be+df \\ ag+ch & bg+hd \end{pmatrix}$

Observe

$ae + bg \neq ae + cf$

$\therefore MN \neq NM$

and / unless $MN = NM = I_n$ where n is the dimension of the square matrices.

1

For $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, the determinant of $A^T A^{-1}$ is;

(a) $\sec^2 x$ (b) $\cos 4x$ (c) 1 (d) 0

How! (5 MARKS) SOLUTION IS IN PAGE 5.!!!

We know $A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$

$\det(A) = (1 \times 1) - (\tan x)(-\tan x)$

$= 1 - (-\tan^2 x)$

$= 1 + \tan^2 x$.

For A, $\text{adj}(A) = \begin{pmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{pmatrix}$

$C_{11} = 1$

$C_{12} = -\tan x$

$C_{21} = \tan x$

$C_{22} = 1$

$\therefore \text{adj}(A) = \begin{pmatrix} 1 & \tan x \\ -\tan x & 1 \end{pmatrix}$

$A^{-1} = \frac{1}{1+\tan^2 x} \begin{pmatrix} 1 & \tan x \\ -\tan x & 1 \end{pmatrix}$

$A^{-1} = \begin{pmatrix} \frac{1}{1+\tan^2 x} & \frac{\tan x}{1+\tan^2 x} \\ \frac{-\tan x}{1+\tan^2 x} & \frac{1}{1+\tan^2 x} \end{pmatrix}$

$A^T = \begin{pmatrix} 1 & -\tan x \\ \tan x & 1 \end{pmatrix}$

$A^T \cdot A^{-1} = \begin{pmatrix} 1 & -\tan x \\ \tan x & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{1+\tan^2 x} & \frac{\tan x}{1+\tan^2 x} \\ \frac{-\tan x}{1+\tan^2 x} & \frac{1}{1+\tan^2 x} \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{1+\tan^2 x} & -\frac{\tan^2 x}{1+\tan^2 x} \end{pmatrix}$

Find A^{-1} for

$$A = \begin{pmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{pmatrix}$$

(5 MARKS)

$\text{adj}(A) = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$

We know that $A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$

using the first column,

$\det(A) = 2\{(5 \times -4) - (-1 \times 2)\} + 1\{(2 \times 3) - (-4 \times -4)\}$

$= 2(-20 + 2) + 1(6 - 16)$

$= -46$

Cofactors

$C_{11} = (-4 \times 5) - (2 \times -1) = -18$

$C_{12} = -[(0 \times 5) - (2 \times 1)] = +2$

$C_{13} = (0 \times -1) - (1 \times -4) = 4$

$C_{22} = (5 \times 2) - (-4 \times 1) = +14$

$C_{21} = -[(2 \times 5) - (-4 \times 1)] = -11$

$C_{23} = -[(2 \times -1) - (3 \times 1)] = +5$

$C_{31} = (3 \times 2) - (-4 \times -4) = -10$

$C_{32} = -[(2 \times 2) - (-4 \times 0)] = -4$

$C_{33} = (2 \times -4) - (0 \times 3) = -8$

$A^{-1} = -\frac{1}{46} \begin{pmatrix} -18 & -11 & -10 \\ 2 & 14 & -4 \\ 4 & 5 & -8 \end{pmatrix}$

$A^{-1} = \begin{pmatrix} \frac{9}{23} & \frac{11}{46} & \frac{5}{23} \\ -\frac{1}{23} & -\frac{7}{23} & \frac{2}{23} \\ -\frac{2}{23} & -\frac{5}{46} & \frac{4}{23} \end{pmatrix}$

The Euclidean norm (length of the vector) $(4, -2, -6)^T$ is; (4 MARKS)

(a) $\sqrt{48}$ (b) $\sqrt{24}$ (c) $\sqrt{12}$ (d) $\sqrt{56}$

$(4,-2,-6)$ ⇒ $= \sqrt{(4)^2+(-2)^2+(-6)^2}$

$= \sqrt{16+4+36}$

$= \sqrt{56}$

7 Let X be a square matrix. Consider the following two statements on X.

I. X is invertible.

II. Determinant of X is non-zero.

Which one of the following is true?

(a) $I \Longrightarrow II : II \nRightarrow I$ (b) $II \Longrightarrow I : I \nRightarrow II$ (c) $I \nRightarrow II : II \nRightarrow I$ (d) I and II are equivalent statements.

Explain! (6 MARKS)

We know that II => I

then $X^{-1} = \frac{1}{\det(X)} \cdot \operatorname{adj}(X)$

if $\det(X) \neq 0$,

when $\det(X) = 0$, $\frac{1}{\det(X)} = \frac{1}{0}$ undefined

and thus we can't get the inverse of X.

We know that I => II.

If $X^{-1} = \frac{1}{\det(X)} \cdot \operatorname{adj}(X)$

because X^{-1} exists, $\det(X) \neq 0$

then thus we can get X^{-1} since $\det(X) \neq$ undefined.

∴ Statements I and II are equivalent statements.

8 The matrix P is the inverse of a matrix Q. If I denotes the identity matrix, which of the following options is correct?

(a) $PQ = I, QP \neq I$ (b) $QP = I, PQ \neq I$ (c) $PQ = I, QP = I$ (d) $PQ = QP = I$

Show! (5 MARKS)

$\begin{pmatrix}1&2\\3&4\end{pmatrix}=5\begin{pmatrix}2&1\\3&4\end{pmatrix}=5\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}2&1\\3&4\end{pmatrix}=5\begin{pmatrix}8&9\end{pmatrix}$

$\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}10&5\\15&10\end{pmatrix}=\begin{pmatrix}40&25\\90&95\end{pmatrix}$

When $PQ: V\to V$, $PQ=I$.

Let $P=Q^{-1}$, and $\vec{x}$ and $\vec{y}$ vectors such that

$Q(\vec{x})=\vec{y}$ and $P(\vec{y})=\vec{x}$.

$PQ(\vec{x})=P(Q(\vec{x}))$

$PQ(\vec{x})=P(\vec{y})$ since $Q(\vec{x})=\vec{y}$,

$PQ(\vec{x})=\vec{x}$ since $P(\vec{y})=\vec{x}$

$\therefore PQ=I$. $\Rightarrow PQ: V\to V$

When $QP: V\to V$, $QP=I$.

Let $P=Q^{-1}$ and $\vec{x}$ and $\vec{y}$ be vectors such that

$Q(\vec{x})=\vec{y}$ and $P(\vec{y})=\vec{x}$.

$QP(\vec{y})=Q(P(\vec{y}))$

$QP(\vec{y})=Q(\vec{x})$ since $P(\vec{y})=\vec{x}$

$QP(\vec{y})=\vec{y}$ since $Q(\vec{x})=\vec{y}$, $\Rightarrow QP: V\to V$

$\therefore QP=I$.

$\therefore PQ=QP=I$.

(3)

(1)

NOTE!

(i.) "Remember to stay in your own row and column during the exam - you are the determinant of your success!"

(ii.) "Don't overthink - this is not Gauss's birthday party. Just row-reduce your stress and focus!"

END

Question 4

$A=\begin{pmatrix}1&\tan x\\-\tan x&1\end{pmatrix}$

$A^T=\begin{pmatrix}1&-\tan x\\\tan x&1\end{pmatrix}$

$A^{-1}=\frac{1}{\det(A)}\cdot adj(A)$

$\det(A)=(1)(1)-(\tan x)(-\tan x)$

$=1+\tan^2 x$

$adj(A)=\begin{pmatrix}C_{11}&C_{21}\\C_{12}&C_{22}\end{pmatrix}$

$C_{11}=1$

$C_{12}=-\tan x$

$C_{21}=\tan x$

$C_{22}=1$

$A^{-1}=\frac{1}{1+\tan^2 x}\begin{pmatrix}1&\tan x\\-\tan x&1\end{pmatrix}$

(2)

$|A^T\cdot A^{-1}|=\left|\begin{pmatrix}1&-\tan x\\\tan x&1\end{pmatrix}\frac{1}{1+\tan^2 x}\begin{pmatrix}1&\tan x\\-\tan x&1\end{pmatrix}\right|$

$=\left|\frac{1}{1+\tan^2 x}\begin{pmatrix}1&-\tan x\\\tan x&1\end{pmatrix}\begin{pmatrix}1&\tan x\\-\tan x&1\end{pmatrix}\right|$

$=\left|\frac{1}{1+\tan^2 x}\begin{pmatrix}1+\tan^2 x&0\\0&1+\tan^2 x\end{pmatrix}\right|$

$=\left(\frac{1}{1+\tan^2 x}\right)(1+\tan^2 x)^2=\frac{1}{1+\tan^2 x}\times 1+\tan^2 x\times 1+\tan^2 x$

$=1+\tan^2 x.$

$=\sec^2 x.$